

Data Processing - SNPs

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Read in DNA sequences

```
alignment <- msa(seqs, method = "ClustalW")
```

Align the sequences

```
## use default substitution matrix
```

```
# Convert to a character matrix
aligned_strings <- as.character(msaConvert(alignment, type = "seqinr::alignment")$seq)

# Then split into individual characters
aligned_matrix <- do.call(rbind, strsplit(aligned_strings, split = ""))

colnames(aligned_matrix) <- 1:ncol(aligned_matrix)
rownames(aligned_matrix) <- names(seqs)
```

```
# Find polymorphic sites (i.e. locations where bases differ across sequences)
variable_sites <- which(apply(aligned_matrix, 2, function(col) length(unique(col)) > 1))

# Print the indices of the polymorphic sites and the corresponding bases for all sequences
aligned_matrix[, variable_sites, drop = FALSE]
```

Find differences among these DNA sequences

```
##      7   13  22  34  40  71  73  85  88  103 106 124 130 136 152 196 208 295
## [1,] "T" "A" "C" "A" "C" "C" "A" "A" "T" "G" "T" "T" "C" "A" "T" "C" "A" "C"
## [2,] "T" "A" "C" "A" "C" "C" "A" "G" "T" "A" "T" "T" "A" "A" "T" "C" "A" "C"
## [3,] "A" "G" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "T" "A" "T"
## [4,] "A" "G" "C" "A" "C" "C" "G" "A" "T" "A" "C" "T" "C" "A" "T" "T" "A" "T"
## [5,] "A" "G" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "T" "A" "T"
## [6,] "A" "G" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "T" "T" "T"
## [7,] "A" "G" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "T" "A" "T"
## [8,] "A" "G" "C" "A" "C" "C" "A" "A" "T" "A" "A" "T" "C" "A" "T" "T" "A" "T"
## [9,] "A" "G" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "T" "A" "T" "T" "A" "T"
## [10,] "A" "A" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "T" "A" "T" "T" "A" "T"
## [11,] "A" "A" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "T" "A" "T"
```

```

## [12,] "A" "A" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "T" "A" "T"
## [13,] "A" "A" "T" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "T" "A" "T"
## [14,] "A" "A" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "C" "A" "C"
## [15,] "A" "A" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "C" "A" "C"
## [16,] "A" "A" "C" "A" "C" "T" "A" "A" "T" "A" "C" "T" "C" "A" "T" "C" "A" "C"
## [17,] "A" "A" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "C" "A" "C"
## [18,] "A" "A" "T" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "C" "A" "C"
## [19,] "A" "A" "C" "A" "C" "C" "A" "A" "T" "A" "C" "C" "C" "A" "T" "C" "A" "C"
## [20,] "A" "A" "C" "A" "C" "C" "A" "A" "T" "A" "C" "T" "C" "A" "T" "C" "A" "C"
## [21,] "A" "A" "C" "G" "T" "C" "A" "A" "C" "A" "A" "T" "C" "A" "C" "C" "A" "C"
## [22,] "A" "A" "C" "G" "T" "C" "A" "A" "C" "A" "A" "T" "C" "G" "C" "C" "A" "C"
##      316 430 442 457 514 546 574
## [1,] "A" "A" "T" "T" "A" "G" "A"
## [2,] "A" "A" "T" "T" "A" "G" "A"
## [3,] "A" "A" "T" "T" "A" "G" "A"
## [4,] "A" "A" "T" "T" "A" "G" "A"
## [5,] "A" "A" "T" "T" "A" "A" "A"
## [6,] "A" "A" "T" "T" "A" "G" "A"
## [7,] "A" "G" "T" "T" "A" "G" "A"
## [8,] "A" "A" "T" "T" "A" "G" "A"
## [9,] "A" "A" "T" "T" "A" "G" "A"
## [10,] "A" "A" "T" "T" "A" "G" "A"
## [11,] "A" "A" "T" "T" "A" "G" "A"
## [12,] "A" "A" "T" "T" "A" "G" "G"
## [13,] "A" "A" "T" "T" "A" "G" "A"
## [14,] "A" "A" "T" "C" "G" "G" "A"
## [15,] "A" "A" "T" "T" "G" "G" "A"
## [16,] "A" "A" "T" "T" "G" "G" "A"
## [17,] "G" "A" "T" "T" "G" "G" "A"
## [18,] "G" "A" "T" "T" "G" "G" "A"
## [19,] "A" "A" "T" "T" "A" "G" "A"
## [20,] "A" "G" "T" "T" "A" "G" "A"
## [21,] "A" "A" "A" "T" "A" "G" "A"
## [22,] "A" "A" "A" "T" "A" "G" "A"

```